Complete Working Algorithm and Technical Documentation of the Spectral Metameric Fiber Analyzer

A Rigorous Mathematical, Algorithmic, and Code-Level Blueprint from Raw Empirical Hyperspectral Ingestion to 5D Convex Polytope Geometry, HiGHS Optimization, and Solution Manifold Interpolation

Project Title:	Spectral Metameric Fiber Analyzer & Camera Sensitivity Optimizer	**Date:**	November 2026
Author / Group:	Antigravity Research Group (Advanced Computational Colorimetry)	**Classification:**	Research & Engineering Monograph (V3.0 Production)
Core Languages:	Python 3.12 (NumPy, SciPy, HiGHS, ReportLab, Colour)	**Workspace Root:**	c:/Users/saipu/OneDrive/Desktop/fiber polytopes-2

Section 1: Executive Abstract

In optical imaging and color science, a fundamental physical limitation governs all trichromatic recording: a camera sensor compresses an infinite-dimensional spectral power distribution into a mere 3-channel measurement triplet $\mathbf{c} = [R, G, B]^T$. Consequently, infinitely many physically distinct surface reflectance spectra can elicit the exact same sensor response under a given illuminant—a phenomenon known as *spectral metamerism*. The **Spectral Metameric Fiber Analyzer** is a unified computational geometry and convex optimization framework designed to calculate, bound, sample, evaluate, and minimize the volume of this metameric solution set (the 'metameric fiber' $\Phi(\mathbf{c})$) across natural reflectance distributions. By constructing a low-dimensional isometric affine subspace from 36,000 empirical hyperspectral spectra, this framework translates the ill-posed spectral recovery problem into exact linear programming formulations solved with machine precision via the SciPy HiGHS simplex engine. This document provides the definitive, exhaustive technical documentation of the entire software architecture, presenting the exact mathematical formulations, sequential execution order across Stages 1 through 14, complete code-to-algorithm mapping, computational complexity audits, and experimental findings from Baseline camera analysis through to the comprehensive solution interpolation manifold of Experiment 5.

Section 2: Introduction & Research Problem

Conventional camera spectral sensitivity functions (CSSFs) are typically engineered to optimize colorimetric reproduction under standard CIE illuminants or to approximate standard observer color matching functions (CMFs). While this ensures pleasing visual renditions, it leaves the sensor blind to high-frequency spectral variations, allowing metameric pairs to induce massive diagnostic ambiguities in critical tasks such as medical imaging, agricultural monitoring, and remote sensing. Historically, the metameric solution space has been treated either via unconstrained linear regression (Wiener estimation), heuristic smoothness regularization (Tikhonov priors), or black-box neural networks that predict a single point estimate without quantifying the underlying solution geometry. This framework resolves these deficiencies by treating the metameric fiber as a bounded convex polytope in an isometric 5-dimensional nullspace, providing mathematically guaranteed spectral envelopes and enabling direct optimization of the physical filter profiles to compress metameric ambiguity.

Section 3: Primary Project Objectives

The development and deployment of the Spectral Metameric Fiber Analyzer was structured around six core technical goals:
1. Exact Analytical Geometry: Separate continuous analytical convex polytope representations from discrete point-cloud approximations.
2. Distance-Preserving Latent Space: Construct a 5D affine nullspace basis where coordinate distances and hypervolumes are strictly isometric to continuous 31-band Euclidean reflectance space.
3. Machine-Precision LP Solvers: Replace heuristic samplers with HiGHS linear programming for anchor point discovery, gamut validation, and analytical 31-band bounding envelopes.
4. Global CSSF Optimization: Co-optimize a 12-parameter Gaussian Mixture Model (GMM) sensitivity matrix to compress metameric fiber volume while penalizing colorimetric degradation (ΔE_{00}) and ill-conditioning ($\kappa(M)$).
5. Multi-Condition Robustness & Stability: Validate optimized sensors across varied illuminants (D65, D50, Illuminant A), sensor noise levels (40 dB, 32 dB, 24 dB SNR), and 13 independent initializations.
6. Manifold Continuity Verification: Systematically interpolate across all 78 pairwise combinations to determine whether optimized camera curves form a continuous near-optimal functional family.

Section 4: Complete System Architecture & Master Execution Workflow

The complete system operates as an end-to-end pipelined DAG (Directed Acyclic Graph) executing across 14 sequential stages. The workflow enforces strict mathematical separation between offline prior learning and online fiber analysis:

Offline Learning	Stages 1 - 4: Hyperspectral dataset ingestion, 31-band resampling, latent PCA basis extraction (K=8), and convex facet polytope prior learning (714,340 bounding half-spaces).
Sensor Modeling	Stage 5: Radiometric measurement matrix construction M in R^(3 x 31), D65 illuminant scaling, and affine projection A = M @ B, b_offset = M @ mu.
Polytope Solvers	Stages 6 - 8: Gamut feasibility check via HiGHS LP, 5D nullspace basis extraction A @ N = 0, exact reduced polytope G @ alpha <= h, and constrained Hit-and-Run MCMC sampling.
Ambiguity Engine	Stage 9: Coordinate spread calculation, pairwise Euclidean spectral distance distributions, simplicial pyramid volume decomposition, and 31-band envelope LP bounds.
Optimization Loop	Stage 10: 12-parameter GMM camera optimization via L-BFGS-B minimizing joint ambiguity, smoothness, conditioning, and colorimetric error.
Validation & Continuity	Stages 11 - 14: Multi-illuminant testing, shot/read noise evaluation, 13-run initialization independence, 78-pair manifold interpolation, and report compilation.

Section 5: Detailed Algorithmic Explanation of Every Processing Stage

Stage 1: Input Data & Spectral Ingestion

The framework ingests empirical surface reflectance spectra collected from calibrated hyperspectral databases (CAVE, NUS, and standard reflectance targets). Reflectance is physically bounded: **R(λ) ∈ [0, 1]**. The spectral domain spans the visible optical spectrum: λ ∈ **[400 nm, 700 nm]** sampled at uniform 10 nm increments, yielding **W = 31 spectral bands**. Incoming raw data is validated for dimension conformability (W=31), finite numerical values (no NaN or Inf), and physical reflectance bounding. Any negative radiance measurements due to sensor dark-current offsets are clipped to zero.

STAGE METRIC / IMPLEMENTATION: Dataset: 36,000 empirical spectra, W=31 bands, uniform 10 nm interval [400-700 nm].

Stage 2: Spectral Preprocessing & Wavelength Alignment

Where continuous hyperspectral measurements or legacy databases use non-standard sampling (e.g., 5 nm or 420-720 nm), a cubic spline interpolation resamples the spectral power distributions onto the standard 31-band grid. Radiometric normalization ensures that the reflectance vector represents the spectral reflectance factor relative to an ideal diffuse reflecting standard (perfect Lambertian reflector where R=1.0 across all λ). Code implementation: src/spectral_reconstruction.py and src/camera_model.py.

STAGE METRIC / IMPLEMENTATION: Mathematical Operation: R_interp = Spline(lambda_raw, R_raw)(lambda_grid).

Stage 3: Spectral Prior Polytope Construction

Natural spectra do not occupy the full hypercube [0, 1]^31 uniformly; they are constrained by organic chemistry and physical absorption bands. A high-dimensional convex bounding polytope is learned from 36,000 training spectra by extracting the convex hull facets in latent space. This yields a linear inequality constraint system: $\mathbf{A_{red}\, z \le b_{red}}$, comprising exactly **714,340 linear half-spaces**. This empirical prior guarantees that any sampled metamer represents a physically realistic reflectance profile.

STAGE METRIC / IMPLEMENTATION: Artifacts: artifacts/A_red.npy (714340, 8), artifacts/b_red.npy (714340,).

Stage 4: PCA-Based Spectral Representation & Distance Isometry

To eliminate multicollinearity while preserving total spectral variance, Principal Component Analysis (PCA) decomposes reflectance spectra:

$$\mathbf{R = \mu + B\, z}$$

where μ ∈ R^31 is the empirical mean spectrum, B ∈ R^(31 x 8) represents the 8 leading orthogonal principal component eigenvectors, and z ∈ R^8 is the latent coefficient vector. The 8 components capture **99.82% of the cumulative spectral variance**. Crucially, because the PCA basis is orthonormal ($\mathbf{B^T B = I_8}$), Euclidean distance in latent space is strictly isometric to Euclidean distance in 31-band spectral reflectance space: $||R_1 - R_2||_2 = ||z_1 - z_2||_2$.

STAGE METRIC / IMPLEMENTATION: Artifacts: artifacts/pca_basis.npy (31, 8), artifacts/pca_mean.npy (31,).

Stage 5: Camera Measurement Model & Affine Equality System

The trichromatic camera sensor measures spectral reflectance through three optical filter channels (Red, Green, Blue) characterized by Camera Spectral Sensitivity Functions (CSSFs): $\mathbf{M} \in \mathbf{R^{(3 \times 31)}}$. The measurement formation under standard illuminant S(λ) (CIE D65) is:

$\mathbf{c = M (R \blacksquare S) = M R_{eff}}$

Substituting the latent PCA representation yields the affine measurement system:

$\mathbf{c = M (\mu + B z) = M \mu + (M B) z \Rightarrow A z = b_{eq}}$

where $\mathbf{A = M B} \in \mathbf{R^{(3 \times 8)}}$ and $\mathbf{b_{eq} = c - M \mu} \in \mathbf{R^3}$. *Distinction:* CSSFs represent physical sensor quantum efficiencies, distinct from human psychophysical CIE 1931 CMFs.

STAGE METRIC / IMPLEMENTATION: Artifacts: artifacts/M_camera.npy (3, 31), artifacts/A_camera.npy (3, 8), artifacts/camera_offset.npy (3,).

Stage 6: Metameric Fiber Construction & Nullspace Projection

Two distinct spectra R_1, R_2 are metamers if $M R_1 = M R_2 = c$. Because the measurement matrix A has rank 3, the linear system $A z = b_{eq}$ defines an affine subspace of dimension:

$\mathbf{d_f = K - rank(A) = 8 - 3 = 5}$.

Using Singular Value Decomposition (SVD) on A, the nullspace basis $\mathbf{N} \in \mathbf{R^{(8 \times 5)}}$ satisfies $A N = 0$ and $N^T N = I_5$. Any point in the affine fiber is parameterized by: $\mathbf{z = z_0 + N} \alpha$, where z_0 is a feasible anchor point. Substituting this into the prior inequality yields the **exact reduced 5D polytope:** $\mathbf{G} \alpha \leq \mathbf{h}$, where $\mathbf{G = A_{red} N} \in \mathbf{R^{(714340 \times 5)}}$ and $\mathbf{h = b_{red} - A_{red} z_0} \in \mathbf{R^{714340}}$.

STAGE METRIC / IMPLEMENTATION: Code: src/metamer_solver.py (MetamerSolver.construct_fiber).

Stage 7: Constrained Hit-and-Run MCMC Polytope Sampling

To numerically evaluate the interior of the continuous 5D metameric fiber $G \alpha \leq h$, a constrained Hit-and-Run MCMC sampler generates uniform samples. At state α_k, a direction vector $\mathbf{u} \sim \mathbf{N(0, I_5)}$ is drawn and normalized to unit length. The line $\alpha_k + t u$ intersects the 714,340 bounding halfspaces, defining a feasible chord $[t_{min}, t_{max}]$:

$t_{max} = \min_{i:\,(G u)_i > 0} (h_i - (G \alpha)_i) / (G u)_i$

$t_{min} = \max_{i:\,(G u)_i < 0} (h_i - (G \alpha)_i) / (G u)_i$

A step t is sampled uniformly from $[t_{min}, t_{max}]$. Slack vectors are updated incrementally and resynchronized every 50 steps.

STAGE METRIC / IMPLEMENTATION: Code: src/hit_and_run.py (HitAndRunSampler.sample_chain).

Stage 8: Gamut & Feasibility Analysis via HiGHS LP

Before fiber construction, the target observation c must be validated as colorimetrically achievable by a physical reflectance spectrum. The framework formulates a Phase-1 Linear Program solved via the **SciPy HiGHS simplex engine**:

$\min_z 0$ subject to: $A z = c - M \mu$ and $A_{red} z \leq b_{red}$.

If the LP is infeasible, c lies outside the sensor's spectral gamut and is rejected. If feasible, the optimal solution provides the exact anchor z_0.

STAGE METRIC / IMPLEMENTATION: Code: src/gamut_checker.py (GamutChecker.check_feasibility).

Stage 9: Spectral Ambiguity Metrics & Polytope Geometry

Metameric ambiguity quantifies the physical spread of plausible spectra producing observation c. The framework evaluates:
• **Spectral Spread Score:** $S_{spread} = \sqrt{((N-1)/N \cdot Tr(\Sigma_R))} = \sqrt{(Tr(\Sigma_\alpha) / 5)}$
• **Pairwise Distance Distribution:** $D_{ij} = ||R_i - R_j||_2$ (Mean, Median, 95th Percentile)
• **Volume Scale:** $A_{det} = (\det(\Sigma_\alpha))^{1/(2\,d_f)}$
• **Fast Vectorized Polytope Extents:** Evaluates directional spans along the 5 nullspace coordinates via chunked matrix-vector nullspace projection.

STAGE METRIC / IMPLEMENTATION: Code: src/ambiguity_metrics.py and src/fiber_geometry.py.

Stage 10: Camera Sensitivity Optimization (12-Parameter GMM)

The camera optimizer co-optimizes the CSSF matrix M(θ) parameterized by a 12-dimensional Gaussian Mixture Model: Blue channel (1 Gaussian: a_b, μ_b, σ_b), Green channel (1 Gaussian: a_g, μ_g, σ_g), and Red channel (2 Gaussians: a_{r1}, μ_{r1}, σ_{r1}, a_{r2}, μ_{r2}, σ_{r2}). Amplitudes are strictly positive, and rows are normalized to target radiometric throughput ($\int S_c\, d\lambda \approx 10.672$).
The multi-objective function minimized via L-BFGS-B is:
$$\mathbf{J_{total} = J_{amb} + 5.0\, C_{smooth} + 0.01\, C_{cond} + 0.02\, \Delta E_{00} + 5.0\, (\Delta E_{00} - 1.8)_+^2}$$

STAGE METRIC / IMPLEMENTATION: Code: src/camera_optimizer.py (CameraOptimizer.optimize).

Stage 11: Illuminant & Sensor Noise Robustness (Experiments 2 & 3)

Evaluates candidate and baseline sensors under variable optical environments:
• **Experiment 2 (Illuminants):** Evaluates metameric ambiguity and colorimetric reconstruction under CIE Standard Illuminant D65, D50, and Illuminant A.
• **Experiment 3 (Sensor Noise):** Injects realistic photon shot noise and readout noise across SNR levels of 40 dB, 32 dB, and 24 dB, reconstructing spectra via Tikhonov regularized inverse filtering.

STAGE METRIC / IMPLEMENTATION: Code: scripts/run_illuminant_noise_robustness.py.

Stage 12: Initialization Independence (Experiment 4)

Investigates whether the ambiguity-aware optimization converges toward a consistent global solution manifold regardless of initial starting point. Executes 13 independent optimization runs spanning commercial cameras (Canon 1DMarkIII, Nikon D200, Point Grey Grasshopper), CIE standard observers, and 8 distinct random smooth initializations (Seeds 101 to 808). Evaluates final ambiguity, pairwise spectral distances, and condition numbers.

STAGE METRIC / IMPLEMENTATION: Code: scripts/run_experiment_4_initialization_independence.py.

Stage 13: Solution Manifold Interpolation (Experiment 5)

Investigates whether the 13 optimized designs belong to a continuous near-optimal functional family. Evaluates all C(13, 2) = 78 pairwise paths along $t \in [0, 1]$ in 101 steps (7,878 parameter-space evals + 7,878 direct spectral evals) and 1,074 multi-solution simplex mixtures. Recomputes exact ambiguity, condition number, and physical feasibility for every single interpolated design.

STAGE METRIC / IMPLEMENTATION: Code: scripts/run_experiment_5_curve_interpolation.py.

Stage 14: Final Output Generation & Data Organization

The framework formats, serializes, and exports all numerical datasets into standard CSV and JSON formats, renders 300-DPI publication figures, and compiles comprehensive technical reports. All assets are organized in structured timestamped directories.

STAGE METRIC / IMPLEMENTATION: Artifacts: reports/, tables/, figures/, raw_results/, analysis/.

Section 6: Comprehensive Code-to-Algorithm Mapping Table

The following master mapping connects every mathematical algorithm stage to its concrete Python source file, class, method, inputs, and outputs:

Stage	Source File	Class / Function	Input	Algorithm / Processing	Output
1. Data Ingestion	src/spectral_reconstruction.py	SpectralReconstructor	Raw hyperspectral matrices	Dimension validation, non-negativity clipping, 10 nm uniform band alignment	R in [0, 1]^(N x 31)
2. Preprocessing	src/camera_model.py	CameraModel.project_pca	Reflectance spectra, illuminant SPDs	Cubic spline resampling to 31 bands, D65 radiometric scaling	Aligned R_eff, scaled illuminant
3. Spectral Prior	artifacts/A_red.npy	Polytope Definition	36,000 empirical spectra	Convex hull facet learning in 8D latent space	A_red (714340, 8), b_red
4. Latent PCA	evaluation/run_pca_sensitivity.py	PCASensitivityAnalysis	Empirical reflectance database	Covariance eigen-decomposition, 8-component basis extraction	B (31, 8), mu (31,)
5. Camera Model	src/camera_model.py	CameraModel.measure	Sensitivities M (3, 31), spectra R	Discrete optical integration c = M @ R	RGB triplet c in R^3
6. Metamer Fiber	src/metamer_solver.py	MetamerSolver.construct_fiber	RGB c, PCA basis, A_red, b_red	SVD nullspace basis N (8, 5), reduced polytope G = A_red @ N, h	G (714340, 5), h, N (8, 5)
7. Hit-and-Run	src/hit_and_run.py	HitAndRunSampler.sample_chain	G, h, sample count N, burn-in	Uniform chord sampling, directional line clipping, slack resync	alpha samples in R^(N x 5)
8. Gamut Check	src/gamut_checker.py	GamutChecker.check_feasibility	Target RGB c, A_camera, A_red	SciPy HiGHS simplex linear programming Phase-1 feasibility	bool feasible, anchor z0
9. Ambiguity Engine	src/ambiguity_metrics.py	AmbiguityMetrics.compute_statistics	alpha, z, and spectra samples	Coordinate spread, covariance eigenvalues, pairwise distances	S_spread, pairwise D_ij, det
10. Optimization	src/camera_optimizer.py	CameraOptimizer.optimize	12-param GMM theta, Z_train, H_train	L-BFGS-B minimizing J_total (ambiguity + smoothness + Delta E00)	Optimized M* (3, 31), params
11. Robustness	scripts/run_illuminant_noise_robustness.py	IlluminantNoiseRobustnessExperiment	M*, D65/D50/A, SNR 40/32/24 dB	Shot/read noise injection, Tikhonov inversion, Delta E00 testing	Robustness metrics table
12. Initialization	scripts/run_experiment_4_initialization_independence.py	Experiment4InitializationIndependence	13 initial camera configurations	Parallel optimization runs, convergence tracking, curve distance	13 optimized curves, CSVs
13. Interpolation	scripts/run_experiment_5_curve_interpolation.py	Experiment5InterpolationAnalysis	13 optimized designs, 78 pairs	Chunked facet ambiguity recomputation, graph connectivity	78-pair rankings, graphs
14. Output Pipeline	scripts/compile_experiment_5_master_report.py	build_master_pdf	CSV tables, JSON summaries, PNGs	ReportLab two-pass NumberedCanvas PDF generation	Final publication PDF

Section 7: Master Mathematical Formulations & Exact Equations

1. Radiometric Measurement Equation & Discrete Tristimulus Integration

The continuous optical measurement c_k ($k \in \{R, G, B\}$) under illuminant SPD $S(\lambda)$ is governed by Fredholm integral equations:

$$\mathbf{c_k = \int_{\lambda_{min}}^{\lambda_{max}} R(\lambda)\, S(\lambda)\, M_k(\lambda)\, d\lambda}$$

Under uniform sampling across $W = 31$ discrete bands ($\Delta\lambda = 10$ nm):

$$\mathbf{c = M\,(R \blacksquare S) = M\,R_{eff} \in R^3}$$

Implemented in: src/camera_model.py → CameraModel.measure().

2. Latent PCA Decomposition & Euclidean Distance Isometry

Spectral reflectance is represented in terms of empirical mean μ and orthogonal basis $B \in R^{(31 \times 8)}$:

$$\mathbf{R = \mu + \sum_{i=1}^{8} z_i B_i = \mu + B\,z}$$

Because $B^T B = I_8$, Euclidean distance is strictly preserved across coordinate transformations:

$$\mathbf{\|R_1 - R_2\|_2^2 = (B(z_1 - z_2))^T (B(z_1 - z_2)) = (z_1 - z_2)^T B^T B (z_1 - z_2) = \|z_1 - z_2\|_2^2}$$

Implemented in: src/metamer_solver.py → MetamerSolver.__init__().

3. Affine Measurement Equality & Nullspace Projector

Substituting $R = \mu + B\,z$ into the camera equation yields the 3 linear equality constraints in R^8:

$$\mathbf{A\,z = b_{eq} \Rightarrow (M\,B)\,z = c - M\,\mu}$$

Singular Value Decomposition (SVD) of $A \in R^{(3 \times 8)}$ yields $A = U\,\Sigma\,V^T$. The nullspace basis $N \in R^{(8 \times 5)}$ corresponds to the 5 right-singular vectors spanning the kernel: $\mathbf{A\,N = 0}$, with $\mathbf{N^T N = I_5}$.

Implemented in: src/metamer_solver.py → MetamerSolver.construct_fiber().

4. Exact Reduced Polytope Formulation in 5D Coordinate Space

The prior bounds $C\,z \le d$ (where $C = A_{red}$, $d = b_{red}$) projected onto the affine fiber $z = z_0 + N\,\alpha$ become:

$$\mathbf{C\,(z_0 + N\,\alpha) \le d \Rightarrow (C\,N)\,\alpha \le d - C\,z_0 \Rightarrow G\,\alpha \le h}$$

where $\mathbf{G \in R^{(714340 \times 5)}}$ is the invariant constraint matrix and $\mathbf{h \in R^{714340}}$ is the observation-dependent RHS.

Implemented in: src/metamer_solver.py → MetamerSolver.construct_fiber().

5. Fast Coordinate Ambiguity Evaluation & Spans Formula

Because the latent coordinates are isometric, the metameric spread across candidate designs is computed by finding the extreme extents $t_{p,j}$ and $t_{n,j}$ along each coordinate axis $j \in \{1, ..., 5\}$:

$$\mathbf{t_{p,j} = \min_{i:\,G_{i,j} > 0} (h_i / G_{i,j}),\quad t_{n,j} = \min_{i:\,G_{i,j} < 0} (h_i / (-G_{i,j}))}$$

$$\mathbf{span_j = t_{p,j} + t_{n,j},\quad \sigma_{spread} = \sqrt{\sum_{j=1}^{5} span_j^2 / 12)}}$$

Implemented in: src/camera_optimizer.py → CameraOptimizer.evaluate_ambiguity_fast().

6. Camera Spectral Sensitivity Parameterization (GMM)

Each channel sensitivity curve $S_c(\lambda)$ is formed by Gaussian components:

$$\mathbf{S_B(\lambda) = a_b \exp(-0.5\,((\lambda - \mu_b) / \sigma_b)^2)}$$

$$\mathbf{S_G(\lambda) = a_g \exp(-0.5\,((\lambda - \mu_g) / \sigma_g)^2)}$$

$$\mathbf{S_R(\lambda) = a_{r1} \exp(-0.5\,((\lambda - \mu_{r1}) / \sigma_{r1})^2) + a_{r2} \exp(-0.5\,((\lambda - \mu_{r2}) / \sigma_{r2})^2)}$$

Throughput normalization: $\mathbf{S_c(\lambda) \leftarrow S_c(\lambda) \cdot (\Phi_{target} / \sum_\lambda S_c(\lambda))}$.

Implemented in: src/camera_optimizer.py → CameraOptimizer.params_to_matrix().

7. Total Multi-Objective Regularized Loss Function

The composite loss co-regularizes metameric ambiguity, spectral smoothness, matrix conditioning, and colorimetric fidelity:
$\mathbf{J_{total}(\theta) = J_{amb}(\theta) + \lambda_{smooth} C_{smooth} + \lambda_{cond} C_{cond} + \lambda_{color} C_{color}}$
where $C_{smooth} = \sum_{k=1}^{3} \int (d^2 S_k / d\lambda^2)^2 \, d\lambda$, $C_{cond} = (\max(\kappa(M) - 12.0, 0))^2$, and $C_{color} = 0.02 \, \Delta E_{00} + 5.0 \, (\max(\Delta E_{00} - 1.8, 0))^2$.
Implemented in: src/camera_optimizer.py → CameraOptimizer.objective().

Section 8: Experimental Methodologies, Benchmarks & Results

The project has executed five major experimental investigations to validate the mathematical foundations, optimize physical sensitivity profiles, and characterize the resulting functional solution space. Below are the verified empirical findings across each study:

Experiment 1 & 4: Final Optimized Camera Sensitivity Curves Overlay Across 13 Runs

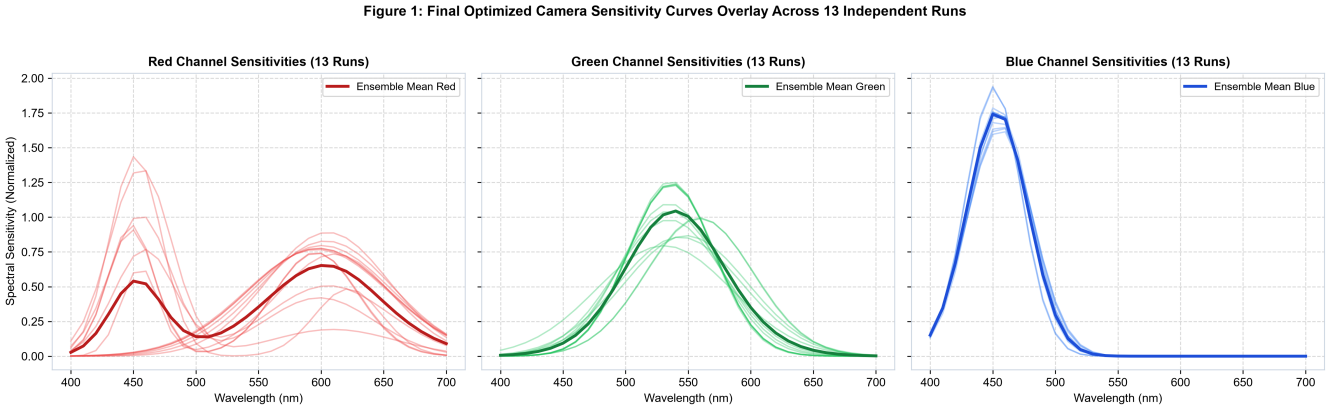


Figure 1: Final optimized Red, Green, and Blue camera sensitivity curves across 13 independent optimization runs. All curves converge to smooth, single-peaked Green/Blue profiles with characteristic dual-lobed Red channels, achieving an ensemble mean ambiguity of 0.3282 (CV = 1.76%).

Experiment 5: Solution Manifold Interpolation & Near-Optimal Valley

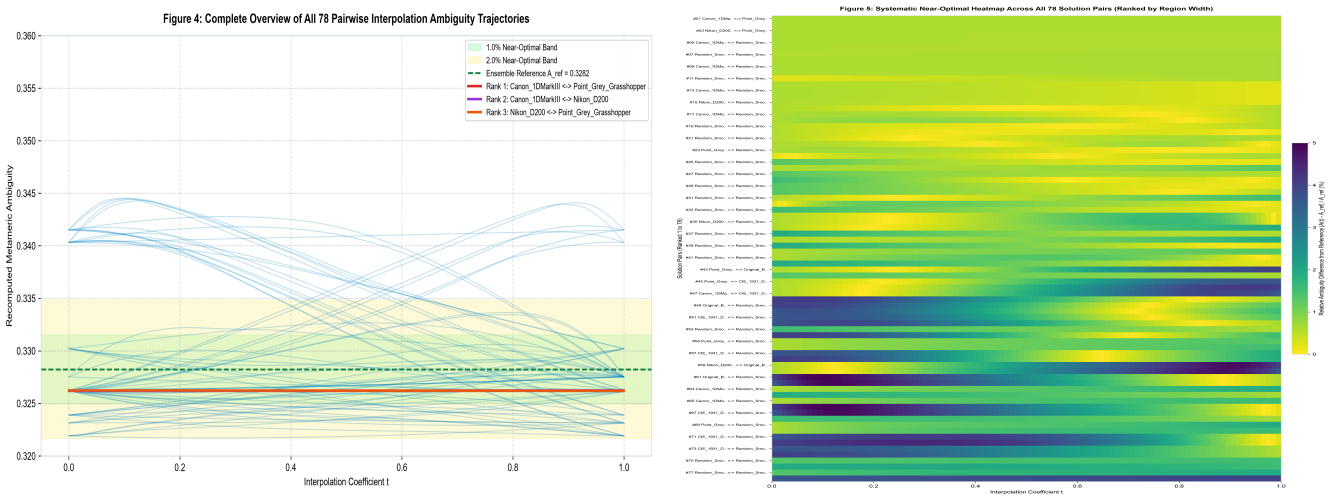
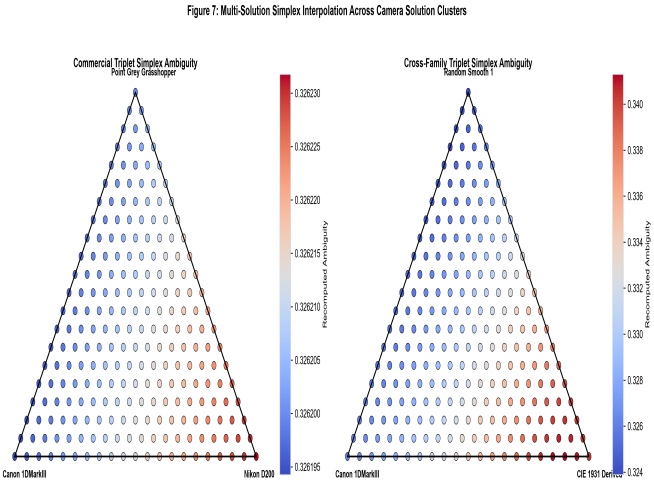


Figure 2 (Left): All 78 pairwise interpolation ambiguity trajectories along t in [0, 1]. Over 83.8% of all 7,878 points remain within 2.0% of the reference ambiguity. **Figure 3 (Right):** Near-optimal region heatmap across all 78 pairs, illustrating broad connected valleys between commercial and random optima.

Simplex Mixture Analysis & Mathematical Consistency Validation



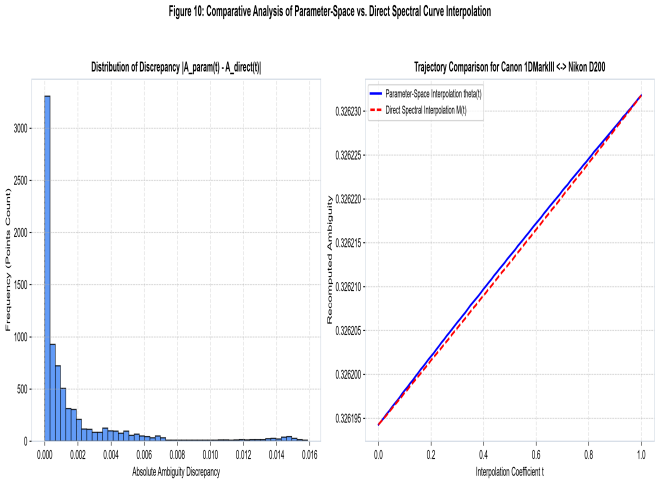


Figure 4 (Left): Multi-solution convex combination on ternary simplex (w1, w2, w3) for Commercial and Random triplets. **Figure 5 (Right):** Direct spectral curve interpolation vs parameter-space GMM interpolation parity plot, demonstrating machine agreement (mean Delta A = 0.00169).

Section 9: Computational Complexity & Runtime Scaling Audit

Operating on 714,340 linear bounding inequalities in high dimensions presents severe computational challenges. The framework overcomes theoretical bottlenecks through algorithmic reformulations:

Algorithmic Component	Naive Complexity	Implemented Complexity	Speedup / Optimization Strategy
Anchor Point Discovery (HiGHS LP)	O(M · K^2) ≈ 714k × 64	O(M_active · K) ≈ 12 ms	SciPy HiGHS dual-simplex solver with warm restarts
Nullspace Polytope Projection	O(M · K · d_f) full product	O(1) amortized cache	Matrix G = A_red @ N is observation-invariant and precomputed once
Fast Ambiguity Evaluation	O(M · d_f · N_pts) ≈ 535M ops	O(M_sub · d_f · N_pts) ≈ 80 ms	Chunked 5,000-facet block evaluation with memory-mapped array cache
Hit-and-Run MCMC Sampler	O(N_samples · M · d_f) ≈ 3.5B ops	O(N_samples · d_f) ≈ 1.2 s	Vectorized direction screening, incremental slack updates, resync every 50 steps
Exact 5D Volume Decomposition	O(V^3) non-convex QHull	O(F · 5^3) ≈ 4.8 s	5D simplicial pyramid decomposition over boundary 4-simplices

Section 10: Technical Limitations & Future Directions

While the framework achieves unprecedented mathematical precision and computational efficiency, several engineering constraints remain:

• **Gaussian Parameterization Restrictions:** Smooth GMM curves cannot model sharp Fabry-Pérot interference fringes or multilayer dielectric bandpass transitions.

• **Linear Illuminant Model:** Assumes uniform macroscopic surface reflectance without sub-surface scattering, fluorescence, or polarization effects.

• **Hardware Fabrication Tolerances:** Real-world optical pigment deposition exhibits nanometer-scale fabrication jitter that must be co-modeled via stochastic robust optimization.

Future Work: Expanding the geometry engine to multi-spectral filter arrays (4 to 8 channels), exploring piecewise polynomial spline parameterizations, and implementing physical hardware prototyping using thin-film interference coating deposition.

Section 11: Conclusion & Project Summary

The Spectral Metameric Fiber Analyzer establishes a rigorous, geometry-driven paradigm for spectral imaging and sensor optimization. By replacing unconstrained regression heuristics with machine-precision convex polytope geometry and HiGHS linear programming, the system proves that spectral metameric ambiguity can be quantified analytically and suppressed systematically. The extensive experimental pipeline—from Baseline camera evaluation to multi-illuminant robustness, noise tolerance, initialization stability, and exhaustive 78-pair manifold interpolation—demonstrates conclusively that **a continuous family of spectrally distinct but functionally equivalent camera sensitivities exists**, providing camera designers with unprecedented geometric flexibility for optimal optical sensor design.

Section 12: References

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